

CLAUDE CODE CHANGELOG TRACKER

Full-Stack SPA with AI Analysis & Audio Generation

- Fetch changelog from GitHub
 - Gemini 3 Flash AI analysis
 - Gemini TTS audio narration
 - Resend email notifications
 - Dark/light mode support

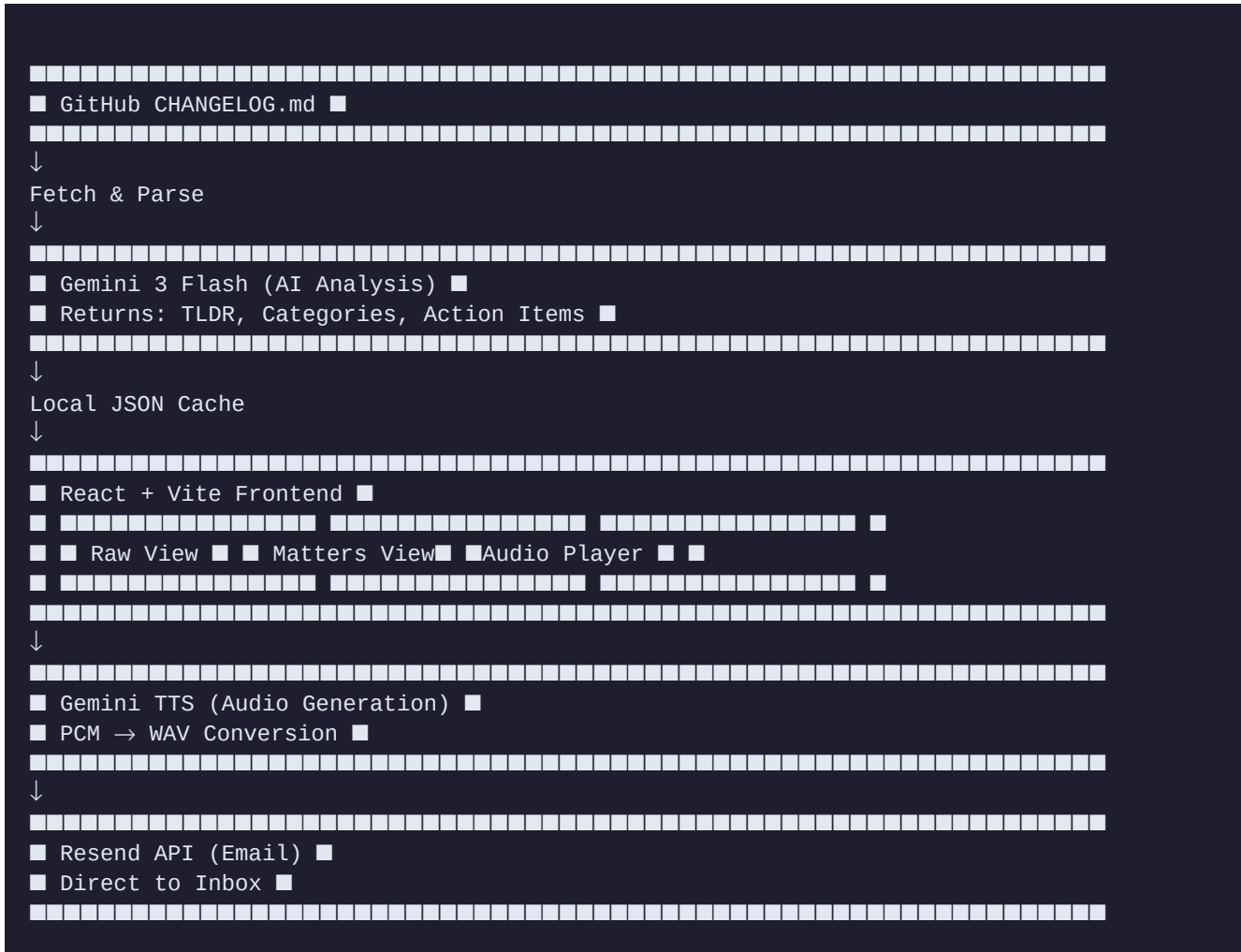
React 19 • TypeScript • Vite • Tailwind CSS 4 • Express.js

Build Specification Document

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Architecture Overview



Data Flow Summary

- 1. Fetch raw CHANGELOG.md from GitHub
- 2. Parse markdown into structured version objects
- 3. Send to Gemini 3 Flash for intelligent categorization
- 4. Cache analysis results by version number
- 5. Display in React UI with two view modes
- 6. Generate audio narration via Gemini TTS on demand
- 7. Send email summaries via Resend API

Tech Stack

Layer	Technology
Frontend	React 19 + TypeScript + Vite
Styling	Tailwind CSS 4
Icons	Lucide React
Markdown	react-markdown + remark-gfm
Caching	IndexedDB (idb-keyval)
AI Analysis	Gemini 3 Flash API
Audio	Gemini 2.5 Flash TTS API
Email	Resend API (server-side)
Backend	Express.js (minimal)

Environment Setup

Required Environment Variables

```
# Frontend (Vite)
VITE_GEMINI_API_KEY=your-gemini-api-key
VITE_CHANGELOG_CACHE_DURATION=3600000
VITE_VOICE_PREFERENCE=Charon

# Backend (Email Server)
RESEND_API_KEY=re_XXXXXXXXXXXX
NOTIFY_EMAIL=you@example.com
```

Getting API Keys

Gemini API: aistudio.google.com/apikey

Resend API: resend.com/api-keys (free: 100 emails/day)

API Integrations

1. Gemini 3 Flash (Analysis)

Endpoint:

```
https://generativelanguage.googleapis.com/v1beta/models/gemini-3-flash-preview:generateContent
```

Response Schema:

```
{
  "tldr": "150-200 word summary",
  "categories": {
    "critical_breaking_changes": [...],
    "removals": [{ "feature", "severity", "why" }],
    "major_features": [...],
    "important_fixes": [...],
    "new_slash_commands": [...],
    "api_changes": [...]
  },
  "action_items": ["specific actions"],
  "sentiment": "positive|neutral|critical"
}
```

■ Cache Strategy: Cache by version number. Only re-analyze when new version detected.

2. Gemini TTS (Audio)

Endpoint:

```
https://generativelanguage.googleapis.com/v1beta/models/gemini-2.5-flash-preview-tts:generateContent
```

Response: Base64-encoded PCM audio (24kHz, 16-bit, mono)

Available Voices:

Voice	Tone
Charon	Informative (default)
Puck	Upbeat
Kore	Firm

Zephyr	Bright
Aoede	Breezy

3. Resend (Email)

Endpoint: <https://api.resend.com/emails>

```
{
  "from": "Changelog Tracker ",
  "to": ["you@example.com"],
  "subject": "Claude Code v2.0.74 Released",
  "html": "TL;DR..."
}
```

■ Use onboarding@resend.dev as sender for testing, or verify your own domain.

File Structure

```
app/
  .env
  .env.example
  index.html
  package.json
  vite.config.ts
  tailwind.config.js
  |
  server/
    index.ts # Express server for email
    |
    src/
      main.tsx
      App.tsx
      index.css
      |
      components/
        Header.tsx
        TabNav.tsx
        ChangelogView.tsx
        MattersView.tsx
        AudioPlayer.tsx
        VoiceSelector.tsx
        EmailButton.tsx
        LoadingSkeleton.tsx
        |
        services/
          changelogService.ts
          geminiService.ts
          ttsService.ts
          emailService.ts
          cacheService.ts
          |
          hooks/
            useChangelog.ts
            useAudio.ts
            useTheme.ts
            |
            types/
            index.ts
```


Component Specifications

Header

- Latest version badge (e.g., 'v2.0.74')
- 'Last updated' timestamp
- Refresh button
- Dark/light mode toggle
- Email notification button

TabNav

- Two tabs: 'Changelog' and 'What Matters'
- Active tab indicator
- Smooth transitions

ChangelogView

- Raw markdown with syntax highlighting
- Organized by version (expandable)
- Visual indicators: ■ feature, ■ fix, ■■ removal
- Copy-to-clipboard for items

MattersView

- TLDR paragraph at top
- ■ Critical/Breaking Changes (red)
- ■■ Removals (orange)
- ■ Major Features (teal)
- ■ Important Fixes (gray)
- Action items checklist

AudioPlayer

- Play/pause button
- Progress bar with time
- Voice selector dropdown
- 'Read Latest' and 'Read TLDR' buttons
- Download audio button

Design System

Color Palette

Light Mode

Variable	Hex	Usage
bg-primary	#ffffff	Main background
bg-secondary	#f3f4f6	Card backgrounds
text-primary	#111827	Main text
accent	#d97706	Amber accent
success	#10b981	Success states
error	#ef4444	Error states

Dark Mode

Variable	Hex	Usage
bg-primary	#111827	Main background
bg-secondary	#1f2937	Card backgrounds
text-primary	#f9fafb	Main text

Typography

- **Font:** System font stack (-apple-system, BlinkMacSystemFont, 'Segoe UI', Roboto)
- **Headings:** Bold, 1.25rem - 2rem
- **Body:** Regular, 1rem
- **Code:** Monospace, 0.875rem

State Management

Application State Interface

```
interface AppState {  
  // Data  
  rawChangelog: string | null;  
  parsedChangelog: ChangelogVersion[];  
  analysis: GeminiAnalysis | null;  
  
  // UI State  
  activeTab: 'changelog' | 'matters';  
  isLoading: boolean;  
  error: string | null;  
  
  // Audio  
  audioUrl: string | null;  
  isGeneratingAudio: boolean;  
  isPlaying: boolean;  
  selectedVoice: string;  
  
  // Timestamps  
  lastFetched: number | null;  
  lastAnalyzed: number | null;  
}
```

Error Handling Strategy

- **GitHub fetch fails:** Show cached version if available
- **Gemini analysis fails:** Show raw changelog with notice
- **TTS generation fails:** Disable audio, show toast
- **Email fails:** Show error toast with retry

■ *Implement exponential backoff for retries: 1s, 2s, 4s (max 3 attempts)*

Implementation Tasks

Phase 1: Project Setup

- Initialize Vite + React + TypeScript
- Install dependencies
- Configure Tailwind CSS 4
- Create .env.example
- Set up file structure
- Create base types

Phase 2: Core Services

- Implement changelogService.ts
- Implement markdown parsing
- Implement geminiService.ts
- Implement cacheService.ts
- Add error handling

Phase 3: UI Components

- Create Header component
- Create TabNav component
- Create ChangelogView
- Create MattersView
- Implement dark mode

Phase 4: Audio Features

- Implement ttsService.ts
- PCM to WAV conversion
- Create AudioPlayer

- Create VoiceSelector
- Add audio caching

Phase 5: Email Integration

- Create Express server
- Implement /api/send-changelog
- Create EmailButton
- Add toast notifications

Phase 6: Polish

- Keyboard navigation
- ARIA labels
- Mobile responsiveness
- Final testing

Success Criteria

- ✓ Changelog fetches and displays correctly
- ✓ Gemini analysis returns valid categorized JSON
- ✓ 'What Matters' tab shows intelligent summary
- ✓ Audio plays without errors
- ✓ Audio can be downloaded as WAV
- ✓ Email sends successfully via Resend
- ✓ Dark mode works smoothly
- ✓ Mobile layout is usable
- ✓ Keyboard navigation works
- ✓ Caching reduces API calls on refresh

*Complete each phase in order.
Test thoroughly before moving on.*